

Meet Radadiya

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Summary

AI Engineer specializing in **real-time Voice AI systems, LLM agent orchestration, and telephony integration**. At Crest Infosystems, built production pipelines including an **AI IVR system** (Twilio + ElevenLabs + LiveKit), a **sub-3s Voice RAG chatbot** with cloned voice, an **AI Conference App** (LiveKit + Deepgram + Groq), and a **Video LipSync pipeline** with voice cloning — all deployed in production. Experienced with **SIP trunking, WebRTC, STT/TTS pipelines, and agentic LLM frameworks**. Fine-tuned Llama 3 using Unsloth + LoRA; used DeepSpeed for inference optimization. AWS Certified ML Engineer (Feb 2026).

Experience

Crest Infosystems Pvt. Ltd.

Jul 2025 – Present

Software Engineer L1 – ML Engineer

Surat, Gujarat

- Built a production **AI Voice IVR System** integrating **Twilio SIP trunking, ElevenLabs TTS, and LiveKit** — handling real-time inbound/outbound conversational call flows with AI agents over WebRTC.
- Engineered a real-time **Agentic Voice RAG Chatbot** — full pipeline from voice input (**Groq Whisper STT**) through **fast-talker/slow-thinker LLM agent** to **XTTS cloned voice synthesis** in **under 3 seconds**; optimized with **DeepSpeed** for low-latency inference.
- Building an **AI Conference Application** (ongoing) with real-time audio/video (**LiveKit**), **Deepgram STT**, **Supabase** backend, and **Groq LLM** for live meeting transcription and intelligent summary generation.
- Built an end-to-end **Video LipSync Pipeline with Voice Cloning** — transcribes via **Faster Whisper**, synthesizes cloned voice via **Chatterbox TTS**, generates lip-synced output using **SoulX-Flashead**; deployed on **AWS**.

Avinashi Group of Companies

May 2024 – Jun 2024

AI Research Intern

Surat, Gujarat

- Fine-tuned **Llama and Mistral** using **LoRA/QLoRA (PEFT)** via Hugging Face Transformers for domain-specific NLP tasks, improving analytical accuracy by **30%** and cutting processing time by **40%**.
- Built data preprocessing pipelines with custom **tokenization** strategies, curating high-quality training datasets that boosted model accuracy by **15%**.

Projects

MediSight – Multimodal Medical Report Generator | Python, Unsloth, Llama 3, LoRA, Groq, Streamlit

[Live Demo](#)

- Fine-tuned **Llama 3** using **Unsloth + LoRA (PEFT)** for multimodal medical image understanding; generates structured clinical PDF reports across 8 diagnostic sections from radiology inputs (X-ray, MRI, CT).
- Deployed as a live **Hugging Face Space**, reducing report generation time by **70%** — demonstrating end-to-end LLM fine-tuning, structured output parsing, and production deployment.

DermaScan AI – Skin Disease Classifier | Python, PyTorch, ResNet, Transfer Learning, Streamlit

[Live Demo](#)

- Built a multi-class skin cancer and disease classifier using **fine-tuned CNNs** on dermoscopy datasets across 10+ conditions; deployed as a live **Hugging Face Space** accessible to healthcare users.

Digital Twin for Manufacturing Optimization | Python, Computer Vision, ML

- Built real-time equipment monitoring with **predictive maintenance ML models**, reducing downtime by **35%** and saving 15L annually via anomaly detection and alerting pipelines.

ImmunoQuest – WBC Classifier | Python, PyTorch, ResNet50, Transfer Learning

- Won 1st place** among 50+ teams; classified White Blood Cells into 5 classes using fine-tuned **ResNet50**, achieving **97% Macro F1 Score** with transfer learning and augmentation strategies.

Technical Skills

Languages & Backend: Python, FastAPI, REST APIs, WebSockets, SQL

Voice AI & Telephony: LiveKit, Twilio (SIP Trunking), ElevenLabs, Deepgram, Faster Whisper, Groq Whisper, XTTS, Chatterbox TTS, WebRTC

LLM & Agentic Systems: LangChain, LlamaIndex, Groq, OpenAI API, Agentic RAG, Prompt Engineering

Model Training & Fine-tuning: PyTorch, Hugging Face Transformers, Unsloth, LoRA/QLoRA (PEFT), DeepSpeed

NLP & ML: Transformer Architectures, BPE Tokenization, Embeddings, Vector DBs (FAISS, ChromaDB), Scikit-learn

Cloud & MLOps: AWS (Certified ML Engineer), Supabase, Docker, Git & GitHub, Hugging Face Spaces

Education

Charotar University of Science and Technology <i>Bachelor of Technology – Artificial Intelligence & Machine Learning (CGPA: 8.91 / 10.00)</i>	May 2026 Anand, Gujarat
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Certifications

• AWS Certified Machine Learning Engineer – Associate	<i>Issued Feb 2026 • Expires Feb 2029</i>
• AWS Certified AI Practitioner	<i>Issued Jan 2026 • Expires Jan 2029</i>
• AWS Academy Graduate – Cloud Developing	<i>Issued Mar 2025</i>
• NVIDIA: Generative AI using Diffusion Models	
• NVIDIA: Building Video AI Applications at the Edge (Jetson Nano)	
• NPTEL: Deep Learning for Natural Language Processing	
• NPTEL: Introduction to Machine Learning	

Leadership

Vice Chair <i>Charotar University of Science and Technology</i>	IEEE Computer Society @ CHARUSAT <i>Feb 2025 – Aug 2025</i>
• Led strategic direction for the IEEE CS branch; organized 8+ technical workshops on AI/ML and cloud computing attended by 300+ students , achieving 25% membership growth.	
Project Lead <i>Charotar University of Science and Technology</i>	AI/ML Club @ CHARUSAT <i>Sep 2024 – Aug 2025</i>
• Directed 4 AI project teams, mentoring 15+ students in LLMs and production ML; organized 5+ GenAI workshops and facilitated 3 industry collaborations, growing membership by 30%.	

Achievements

• Kaggle Competition – 1st Place (2025): Won ImmunoQuest ML competition (CHARUSAT) for best WBC classification model among 50+ participating teams .
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